MPhil Thesis MT Summary

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Introduction

What is the Affordable Care Act (ACA)?

- Most comprehensive US health reform since Medicare and Medicaid.
- Expanded access to health insurance and reduced financial barriers to care.
- Motivated by high uninsured rates and substantial gaps in Medicaid eligibility across states.
- Introduced subsidised insurance marketplaces for individuals without employer or public coverage.
- Imposed consumer protection rules on insurers including bans on pre-existing condition exclusion and mandating coverage of essentials.
- Intended to improve coverage stability for low-income individuals.

What changed in Medicaid?

Before the ACA:

- Eligibility required low income and belonging to a specific category (e.g., pregnant, disabled, parent of dependent child).
- Income limits varied by state and were often below the Federal Poverty Level. (e.g., Texas threshold was 17% of the FPL)
- Most low-income adults without dependent children were not eligible.

After the ACA:

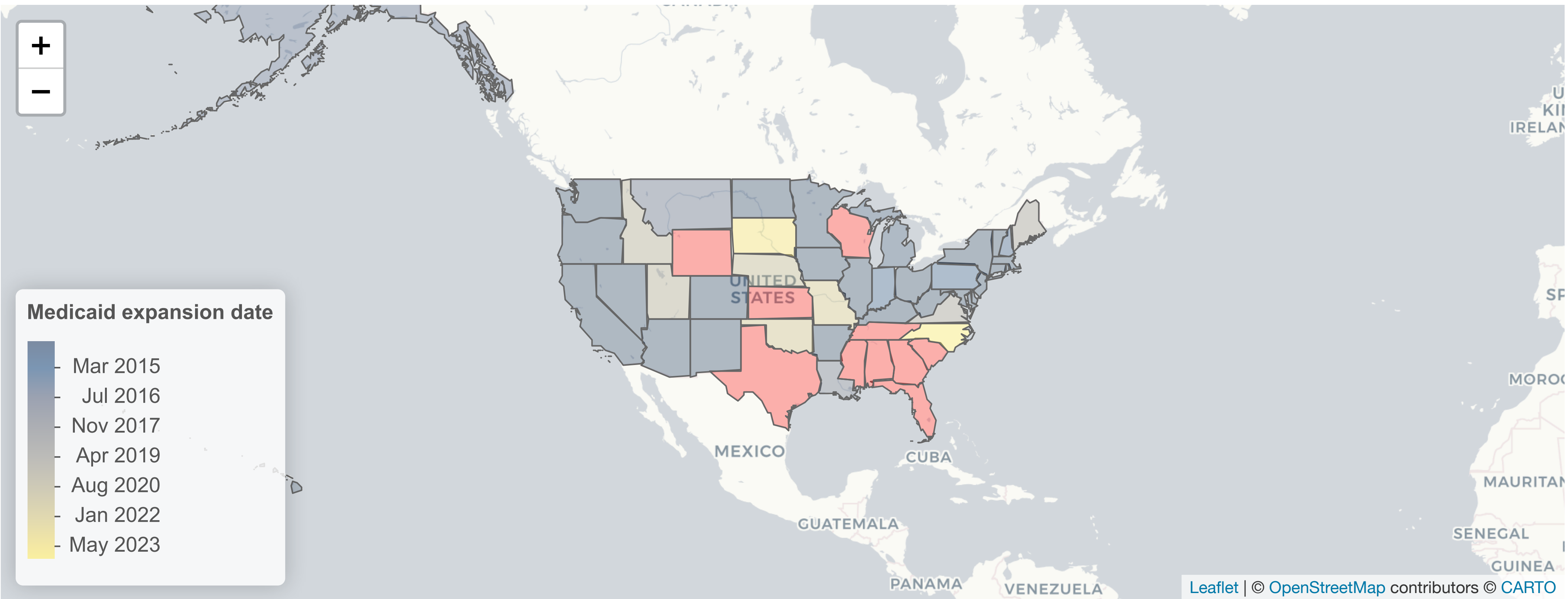
- Eligibility based primarily on income.
- Threshold set at approximately 138% of the Federal Poverty Level.

Medicaid eligibility

Feature	Before ACA	After ACA
Basis for eligibility	Low income and categorical requirement (pregnant, disabled, parent etc.)	Income-based
Income threshold	Varied by state, often below 100% FPL	~138% FPL
Coverage for low-income adults without children	Generally not eligible	Eligible if income ≤138% FPL
State variation	Significant variation	Reduced variation due to optional expansion
Benefit structure	Medicaid covered prescription drugs but	Coverage aligned with essential health benefit

How was the policy rolled out

- **2010:** ACA enacted
- **2012:** Medicaid expansion becomes state-optional
- **2014:** States begin to adopt expansion



Literature Review

What has the current literature investigated?

Topic	Findings	Papers
Insurance coverage & disparities	Large coverage gains concentrated in expansion states; racial/ethnic coverage gaps narrowed.	(Courtemanche et al. 2020; Frean, Gruber, and Sommers 2017; Miller and Wherry 2019)
Access to care	Increased affordability; decreased cost barriers; longer waits and delays	(Miller and Wherry 2017)
Health outcomes	Improvements in self-reported health; reduced adult mortality in expansion states.	(Miller, Johnson, and Wherry 2021; Miller and Wherry 2017)

Topic	Findings	Papers
Financial outcomes	Declines in medical debt and fewer bankruptcies; improved credit scores.	(Brevoort, Grodzicki, and Hackmann 2017; Hu et al. 2018)
Labour market	No significant changes in labour market outcomes; disincentives for labour supply; no change in physician hours.	(Duggan, Goda, and Jackson 2019; Dague, DeLeire, and Leininger 2017; Neprash et al. 2021)

Unexplored area?

Pharmaceutical innovation induced by the Medicaid expansion !

Only one paper investigates this: Garthwaite, Sachs, and Stern (2021)

1. Do the ACA-induced market size expansion for pharmaceuticals cause an increase in pharmaceutical innovation?
2. Is there any evidence that firms shifted R&D effort toward conditions more affected by Medicaid expansion?

Findings:

1. No clear change in clinical trial activity
   - For conditions above the median Medicaid market share (ex-ante demand)
   - For top quartile of conditions which faced coverage increases (ex-post demand)
2. Overall: innovation outcomes did not respond to increases in Medicaid covered market size

What are the gaps?

Gap	Issue in Paper	Proposed Solution
Short time window	Data up to 2016	Extend analysis through 2025
No analysis of firm behaviour reallocation	Paper focuses on innovation volume, not whether firms shifted R&D across therapeutic areas	Investigate if firms pivot towards profitable drug classes
Medicaid rebate system	Does not study interaction of rebate system and innovation incentives	Separate effects of medicaid monopsony power and increased demand on innovation

Research Questions:

Aim: Examine how insurance expansion and pricing rules affect pharmaceutical innovation to guide the design of effective policy frameworks.

1. Does the Medicaid expansion under the ACA affect pharmaceutical innovation over the long run?
2. Do pharmaceutical firms reallocate R&D effort across therapeutic areas following Medicaid expansion?
3. How do Medicaid pricing rules influence incentives to innovate?

Competition within the Pharmaceutical Industry

Paper	Findings
Regan (2008)	After initial generic entry, multiple generics enter and branded–generic price gaps widen. Strong post-patent price competition, with substantial price declines. Observes Market segmentation between price-sensitive and less price-sensitive users.
Grabowski and Kyle (2007)	More profitable products attract more generic entrants and have shorter effective exclusivity. Generic competition has become more intense, with many markets seeing rapid erosion of branded share.
Kyle (2006)	Entry of new drugs into national markets depends on market size and existing competition. Firms are likelier to launch in larger, more profitable markets even when there are incumbents.

Within Class competition

Drugs within the same therapeutic class face strong competitive pressure, as payers view them as close substitutes and actively reallocate utilisation across on-patent alternatives.

Mechanism	Source
Clinically similar drugs constrain each other's pricing and market share.	OECD (2018)
New entrants with small improvements erode incumbent market power.	OECD (2018)
Physicians switch within class based on factors such as efficacy, side-effects, guidelines, dosing.	OECD (2018)

Pipeline competition

Firms invest in R&D to enter profitable disease areas or to displace incumbents with improved drugs.

Mechanism	Source
Entry decisions depend on expected profits and existing competitors	Acemoglu and Linn (2004), Dubois et al. (2015)
Follow-on entry is driven by racing behaviour: multiple firms begin development before the first-in-class drug is approved, aiming to introduce improved variants and capture future market share.	DiMasi and Paquette (2004)
Follow-on drugs intensify competition by weakening the first-mover's market power and giving payers alternative options to negotiate lower prices.	OECD (2018)

Medicaid's Preferred Drug List

Key facts

Rebates:

- Brand drugs: ≥ 23.1 % rebate on average manufacturer price (AMP) or best price
- Generics: ≥ 13 % of AMP
- Inflation rebate: added if prices rise faster than inflation

Preferred Drug Lists:

- States designate PDLs with negotiated rebates, with some firms adopting a uniform PDL to increase bargaining power.

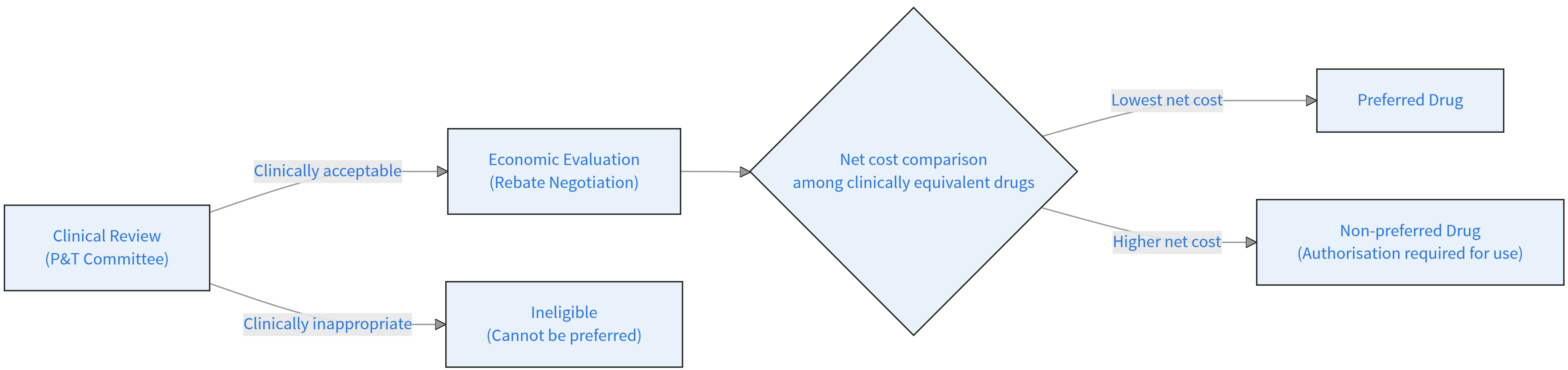
Supplemental Rebate Pools:

- States join multi-state alliances to collectively negotiate deeper rebates.

Medicaid's Preferred Drug List (PDL)

A drug becomes preferred when it is clinically acceptable **and** offers the lowest net cost among substitutes.

1. P&T committee evaluates safety, efficacy and literature on drug.
2. States compare net price among clinically equivalent drugs, with price determining which eligible drugs become most preferred.



States can list multiple preferred drugs when they are not clinically equivalent.

Staying on the PDL

1. Remaining clinically appropriate

- No concerns relating to being unsafe or outdates
- No concerns suggesting it’s less effective than other drugs

2. Remaining cost competitive

- The drug must remain as one of the lowest net-cost options among clinically appropriate drugs
- If a competitor offers a deeper rebate or utilisatioin costs change unfavourably then a drug may be taken off the PDL.

Do R&D incentives exist for firms on the PDL?

Preferred status is not guaranteed.

Firms can only use R&D to increase the clinical value of their drug:

- Show superior outcomes / fewer side-effects
- Improve adherence or reduced downstream costs
    - e.g. once-daily extended-release instead of twice-daily
- Produce better formulations
    - e.g. pen injector instead of vial administration
- Target specific patient subgroups
    - e.g. better outcomes in patients with renal impairment

However, a firm **cannot** innovate their way into a lower production cost

R&D incentives for non-PDL firms

Non-preferred status limits utilisation, firms innovate to escape the prior-authorisation barrier and win preferred placement.

Firms with no existing products have strong incentives to enter a profitable therapeutic class.

Competition methods:

- Enter with a drug with lower costs of production allowing for a larger rebate than incumbents.
- Develop a drug with better efficacy, or improved dosing.
- Create a drug beneficial for patients not well-served by existing preferred products.

Data

Available Data

A large issue with cleaning the data comes from cleaning the datasrets in such a way they can be mapped to the Medicaid SDUD dataset.

This involves mapping each of the datasets to a therapeutic area, as well as finding a method to match firms since firms aren’t even mentioned in the SDUD data. - Standard cleaning and fuzzy matching did not work !

Outcome Data	Stage in innovation pipeline	Processed
Patents	early stage	✅
Clinical trials	mid stage	❌
Drug Approvals	end stage	✅

Medicaid State Drug Utilisation Data

Variable	Info
NDC	11 digit code uniquely identifying drugs, their firm and the package sizing. This is split into three variables: **labeler code** (firm);**product code** (drug for said firm); **package size**
Drug name	Name of the drug
Units reimbursed	Number of units paid for by Medicaid during the Year-Q
Number of prescriptions	Count of prescriptions for a given drug in in that state and Year-Q
Reimbursement amount	Tells us the total amount paid for a given drug in that state and Year-Q. This is split into three variables: total, medicaid and non-medicaid.

Most firms have multiple NDC labeler codes so matching on labeler code is insufficient and further cleaning was required.

Aggregation

Aggregated at broad therapeutic level. - ATC level 1: main anatonmical groups - ATC level 2: further refines each level1 cat into a therapeutic subgroup e.g. diabetes, antibacterials, beta-blockers

ATC Code	Therapeutic Area
A	Alimentary tract and metabolism
B	Blood and blood-forming organs
C	Cardiovascular system
D	Dermatologicals
G	Genito-urinary system and sex hormones
H	Systemic hormonal preparations
J	Anti-infectives for systemic use
L	Antineoplastic and immunomodulating agents

ATC Code	Therapeutic Area
M	Musculo-skeletal system
N	Nervous system
P	Antiparasitic products, insecticides and repellents
R	Respiratory system
S	Sensory organs
V	Various

Collapsing SDUD

After cleaning Medicaid SDUD data it's collapsed to obtain the number of prescriptions and reimbursement amounts. This is used to build demand proxies for a given therapeutic area.

```

# Collapse SDUD data to firm-year-quarter-ATC level

SDUD_firm_ATC <- SDUD_firm_mapped |>
  group_by(
    state,
    firm,
    labeler_code,
    year,
    quarter,
    ATC_Level1,
    ATC_Level2
  ) |>
  summarise(
    prescriptions = sum(number_of_prescriptions, na.rm = TRUE),
    medicaid_reimbursed = sum(medicaid_amount_reimbursed, na.rm = TRUE),
    non_medicaid_reimbursed = sum(non_medicaid_amount_reimbursed, na.rm = T
```

Collapsing Patents

```r

pharma_patents_collapsed <- pharma_patents_full |>
  filter(patent_year >= 2005) |>
  group_by(
    ndc_firm_id,
    firm,
    patent_year,
    patent_quarter,
    ATC1
  ) |>
  summarise(
    num_patents = n(),
    .groups = "drop"
  )
```

Collapsing Drug approvals

```r

fda_approvals_panel <- fda_atc_expanded |>
  group_by(
    sponsor_name,
    labeler_code,
    approval_year,
    approval_quarter,
    atc_level1,
    atc_level2
  ) |>
  summarise(
    n_approvals = n(),
    .groups = "drop"
  )
```

Preliminary Results

RQ1: Response to market size

Four models:

- Ex-ante & FEOLS
- Ex-ante & negative binomial
- Ex-post & FEOLS
- Ex-post & negative binomial

Patents

	FEOLS: Ex Ante	FEOLS: Ex Post	NB: Ex Ante	NB: Ex Post
patent_year = 2005 × d_ex_ante	-31343.962***		0.291	
	(4669.415)		(0.487)	
patent_year = 2006 × d_ex_ante	-23451.153***		0.619	
	(3954.343)		(0.342)	
patent_year = 2007 × d_ex_ante	-18840.349***		1.304***	
	(3727.857)		(0.152)	
* p < 0.05, ** p < 0.01, *** p < 0.001				

	FEOLS: Ex Ante	FEOLS: Ex Post	NB: Ex Ante	NB: Ex Post
patent_year = 2008 × d_ex_ante	-15251.249***		0.961**	
	(3565.934)		(0.280)	
patent_year = 2009 × d_ex_ante	-6864.642		1.267***	
	(3588.229)		(0.253)	
patent_year = 2010 × d_ex_ante	6736.960*		0.917**	
	(2698.481)		(0.220)	
patent_year = 2011 × d_ex_ante	-3260.689		0.500*	
	(1672.657)		(0.179)	
* p < 0.05, ** p < 0.01, *** p < 0.001				

	FEOLS: Ex Ante	FEOLS: Ex Post	NB: Ex Ante	NB: Ex Post
patent_year = 2012 × d_ex_ante	-778.379		0.349**	
	(1171.574)		(0.110)	
patent_year = 2014 × d_ex_ante	-3680.996*		-0.081	
	(1625.624)		(0.151)	
patent_year = 2015 × d_ex_ante	-11177.262***		0.165	
	(2055.870)		(0.318)	
patent_year = 2016 × d_ex_ante	-22804.452***		-0.203	
	(2026.809)		(0.422)	
* p < 0.05, ** p < 0.01, *** p < 0.001				

	FEOLS: Ex Ante	FEOLS: Ex Post	NB: Ex Ante	NB: Ex Post
patent_year = 2017 × d_ex_ante	-20779.892***		0.015	
	(2696.631)		(0.546)	
patent_year = 2018 × d_ex_ante	-23007.524***		0.301	
	(3064.280)		(0.619)	
patent_year = 2019 × d_ex_ante	-31005.461***		-0.316	
	(2728.868)		(0.621)	
patent_year = 2020 × d_ex_ante	-33426.419***		-0.113	
	(2888.516)		(0.594)	
* p < 0.05, ** p < 0.01, *** p < 0.001				

	FEOLS: Ex Ante	FEOLS: Ex Post	NB: Ex Ante	NB: Ex Post
patent_year = 2021 × d_ex_ante	-33182.040***		0.223	
	(2967.989)		(0.517)	
patent_year = 2022 × d_ex_ante	-37354.270***		-0.061	
	(3441.583)		(0.683)	
patent_year = 2023 × d_ex_ante	-39396.839***		0.121	
	(3652.932)		(0.588)	
patent_year = 2024 × d_ex_ante	-39688.931***		0.011	
	(3448.992)		(0.905)	
* p < 0.05, ** p < 0.01, *** p < 0.001				

	FEOLS: Ex Ante	FEOLS: Ex Post	NB: Ex Ante	NB: Ex Post
patent_year = 2025 × d_ex_ante	-45889.451***		-1.617*	
	(5391.641)		(0.703)	
patent_year = 2005 × d_ex_post		-29907.139***		0.241
		(4078.053)		(0.406)
patent_year = 2006 × d_ex_post		-22126.409***		0.632*
		(4138.049)		(0.255)
patent_year = 2007 × d_ex_post		-17534.120***		1.322***
		(3664.484)		(0.228)
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$				

	FEOLS: Ex Ante	FEOLS: Ex Post	NB: Ex Ante	NB: Ex Post
patent_year = 2008 × d_ex_post		-13938.650**		0.999***
		(3631.556)		(0.153)
patent_year = 2009 × d_ex_post		-5822.574		1.327***
		(3203.895)		(0.243)
patent_year = 2010 × d_ex_post		6477.470**		0.855***
		(2086.306)		(0.199)
patent_year = 2011 × d_ex_post		-2765.373		0.514*
		(1508.137)		(0.173)
* p < 0.05, ** p < 0.01, *** p < 0.001				

	FEOLS: Ex Ante	FEOLS: Ex Post	NB: Ex Ante	NB: Ex Post
patent_year = 2012 × d_ex_post		-762.255		0.340**
		(1200.183)		(0.105)
patent_year = 2014 × d_ex_post		-2836.085		0.006
		(1800.325)		(0.139)
patent_year = 2015 × d_ex_post		-9971.610***		0.249
		(1840.704)		(0.224)
patent_year = 2016 × d_ex_post		-21065.105***		-0.086
		(2343.661)		(0.295)
* p < 0.05, ** p < 0.01, *** p < 0.001				

	FEOLS: Ex Ante	FEOLS: Ex Post	NB: Ex Ante	NB: Ex Post
patent_year = 2017 × d_ex_post		-18926.900***		0.172
		(2934.544)		(0.383)
patent_year = 2018 × d_ex_post		-21020.478***		0.459
		(3367.873)		(0.411)
patent_year = 2019 × d_ex_post		-28995.039***		-0.172
		(3520.764)		(0.459)
patent_year = 2020 × d_ex_post		-31600.405***		-0.095
		(3872.419)		(0.490)
* p < 0.05, ** p < 0.01, *** p < 0.001				

FDA Approvals

	FEOLS: Ex Ante	FEOLS: Ex Post	NB: Ex Ante	NB: Ex Post
approval_year = 2005 × d_ex_ante	-165.865***		-0.853	
	(28.961)		(0.594)	
approval_year = 2006 × d_ex_ante	-38.967		-0.904	
	(48.914)		(0.468)	
approval_year = 2007 × d_ex_ante	-46.194		0.327	
	(44.179)		(0.791)	

* p < 0.05, ** p < 0.01, *** p < 0.001

	FEOLS: Ex Ante	FEOLS: Ex Post	NB: Ex Ante	NB: Ex Post
approval_year = 2008 × d_ex_ante	8.070		0.760	
	(31.862)		(0.614)	
approval_year = 2009 × d_ex_ante	106.429*		1.171*	
	(45.759)		(0.507)	
approval_year = 2010 × d_ex_ante	24.171		0.336	
	(28.503)		(0.393)	
approval_year = 2011 × d_ex_ante	6.207		0.163	
	(54.837)		(0.521)	
* p < 0.05, ** p < 0.01, *** p < 0.001				

	FEOLS: Ex Ante	FEOLS: Ex Post	NB: Ex Ante	NB: Ex Post
approval_year = 2012 × d_ex_ante	29.273		0.616*	
	(17.523)		(0.280)	
approval_year = 2014 × d_ex_ante	-48.201***		-0.545*	
	(8.190)		(0.184)	
approval_year = 2015 × d_ex_ante	-0.114		-0.508	
	(21.254)		(0.317)	
approval_year = 2016 × d_ex_ante	43.498		-0.733**	
	(39.364)		(0.243)	
* p < 0.05, ** p < 0.01, *** p < 0.001				

	FEOLS: Ex Ante	FEOLS: Ex Post	NB: Ex Ante	NB: Ex Post
approval_year = 2017 × d_ex_ante	169.714		-0.255	
	(80.894)		(0.374)	
approval_year = 2018 × d_ex_ante	106.348		-0.482	
	(65.745)		(0.314)	
approval_year = 2019 × d_ex_ante	69.224		-0.292	
	(44.310)		(0.216)	
approval_year = 2020 × d_ex_ante	97.360**		-0.479*	
	(28.485)		(0.214)	
* p < 0.05, ** p < 0.01, *** p < 0.001				

	FEOLS: Ex Ante	FEOLS: Ex Post	NB: Ex Ante	NB: Ex Post
approval_year = 2021 × d_ex_ante	10.981		-0.656*	
	(24.570)		(0.269)	
approval_year = 2022 × d_ex_ante	41.538		-0.350	
	(26.852)		(0.236)	
approval_year = 2023 × d_ex_ante	118.083**		-0.362	
	(34.673)		(0.329)	
approval_year = 2024 × d_ex_ante	22.487		-0.391	
	(49.672)		(0.306)	

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

	FEOLS: Ex Ante	FEOLS: Ex Post	NB: Ex Ante	NB: Ex Post
approval_year = 2025 × d_ex_ante	-28.988		-0.372	
	(63.451)		(0.502)	
approval_year = 2005 × d_ex_post		-145.269***		-0.732
		(21.525)		(0.519)
approval_year = 2006 × d_ex_post		-14.801		-0.664
		(32.755)		(0.339)
approval_year = 2007 × d_ex_post		-53.892		0.164
		(29.035)		(0.633)
* p < 0.05, ** p < 0.01, *** p < 0.001				

	FEOLS: Ex Ante	FEOLS: Ex Post	NB: Ex Ante	NB: Ex Post
approval_year = 2008 × d_ex_post		20.682		0.781
		(20.827)		(0.591)
approval_year = 2009 × d_ex_post		114.880***		1.153*
		(19.496)		(0.454)
approval_year = 2010 × d_ex_post		14.510		0.264
		(19.261)		(0.335)
approval_year = 2011 × d_ex_post		28.382		0.332
		(28.963)		(0.394)
* p < 0.05, ** p < 0.01, *** p < 0.001				

	FEOLS: Ex Ante	FEOLS: Ex Post	NB: Ex Ante	NB: Ex Post
approval_year = 2012 × d_ex_post		34.886**		0.619*
		(10.561)		(0.275)
approval_year = 2014 × d_ex_post		-46.164***		-0.499**
		(6.043)		(0.156)
approval_year = 2015 × d_ex_post		-1.481		-0.454
		(19.053)		(0.274)
approval_year = 2016 × d_ex_post		30.414		-0.672**
		(33.122)		(0.199)
* p < 0.05, ** p < 0.01, *** p < 0.001				

	FEOLS: Ex Ante	FEOLS: Ex Post	NB: Ex Ante	NB: Ex Post
approval_year = 2017 × d_ex_post		134.793		-0.291
		(67.329)		(0.310)
approval_year = 2018 × d_ex_post		84.054		-0.459
		(55.940)		(0.268)
approval_year = 2019 × d_ex_post		40.992		-0.368*
		(23.625)		(0.145)
approval_year = 2020 × d_ex_post		89.649**		-0.399*
		(28.607)		(0.171)
* p < 0.05, ** p < 0.01, *** p < 0.001				

RQ2: Reallocation Across Therapeutic Areas

	Patents	FDA Approvals
(Intercept)	0.242***	0.312***
	(0.002)	(0.022)
medicaid_present	-0.111***	-0.136***
	(0.005)	(0.024)
post	0.043***	0.024
	(0.002)	(0.021)
medicaid_present × post	-0.024***	-0.037
	(0.005)	(0.023)
Num.Obs.	108159	3822
R2	0.018	0.083
* p < 0.05, ** p < 0.01, *** p < 0.001		

	Patents	FDA Approvals
R2 Adj.	0.018	0.083
AIC	37554.0	747.1
BIC	37592.3	772.1
RMSE	0.29	0.27
Std.Errors	by: firm	by: sponsor_name
* p < 0.05, ** p < 0.01, *** p < 0.001		

Interpretation of results

1. Significant reduction in patent applications post ACA
2. No significant effect on FDA approvals post ACA
3. Pharma firms reallocate their share of innovation towards diseases which they don’t cover for Medicaid.

Why did innovation respond negatively to Medicaid expansion?

1. Medicaid leads to a higher market size but rebates erode profits
    - Firms on the PDL lose incentives to innovate if they gain net-zero profit from drug development.
2. A firm on the PDL has captured the market
    - A dominant firm will not experience a rise in demand from innovation.
    - Investment in innovation to receive the same level of demand would yield less profits compared to a “do-nothing” approach.

Both of these outweigh the benefits of innovating under a larger market size

Is this consistent with the literature

Results could explain findings in Garthwaite, Sachs, and Stern (2021)!

- Their paper sees no significant effect in clinical trials activity post ACA.
- Clinical trials are a late stage outcome and they only use data for two years post treatment
- We only see an effect on patents - the earliest stage outcome in the pharma-innovation pipeline
- We do not observe an effect on the latest stage indicator even after 10 years

This could suggest that innovation is inelastic in the short-run for longer-term indicators.

- Reviewing updated clinical trials data can shed light to this theory

Next steps

Theoretical model

Two-firm game involving oligopolistic competition:

1. Expansion of Medicaid as part of ACA is a shock that leads to a jump in market demand
2. Firms innovate to capture market demand
3. However, being on PDL decides whether a firm captures the market or not.
4. How do incentives change and why if a firm captures the market vs if it doesn’t.
5. P&T review drugs each year so a drug has the possibility of being removed from the market.

Steps for Hilary Term

1. Investigate failure of parallel trends to confirm validity of results
2. Clean clinical trials data
3. Obtain Medicaid PDL lists
4. Investigate data on price-rebate intensity
5. Investigate RQ3
6. Build two-player game to explain results

Bibliography

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